BunBau v7.1 Comprehensive Implementation Plan
Airtight Live-vs-Demo Zero-Bleed Data Partitioning & Universal Canonical Container UX Standardization Architecture

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Executive Metadata & Mandatory Tracking
> *"You also need to update the application version/build/architecture version for every iteration build."*

- **Milestone Designation:** `v7.1`
- **Current App Version:** `2.7.12 (Build 127)`
- **Target App Version:** `2.7.13 (Build 128)`
- **Target Architecture Version:** `v2.9.6 Airtight Live-vs-Demo Zero-Bleed Data Partitioning & Universal Canonical Container UX Standardization Architecture`
- **Target Status:** **Pending User Approval**

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1. Deep Forensic Codebase Audit: Demo Data Leaks in Live Mode

A comprehensive, code-level forensic audit was conducted across the entire BunBau codebase (`src/lib/tax/`, `src/lib/rag/`, `src/components/tax/`, `src/contexts/`, and `src/app/api/`).

We identified **eight distinct architectural vectors** where synthetic demo data (Alex & Jordan Taylor, 742 Evergreen Terrace, Pacific Cloud Systems Inc., and mock tax facts) leaked, fell back, or bled into Live User Mode:

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┌──────────────────────────────────────────────────────────────────────────────────────────┐
│                    FORENSIC AUDIT: 8 DEMO DATA INFILTRATION VECTORS                      │
├──────────────────────────────────────────────────────────────────────────────────────────┤
│ 1. TaxProfileHelper.ts (Line 151 & 200)                                                  │
│    • resolveLiveTaxpayerProfile() initialized using { ...DEMO_TAX_FILING_PROFILE }       │
│    • Inherited demo employer ('Pacific Cloud Systems Inc.'), demo address, & demo DOBs   │
│                                                                                          │
│ 2. HybridRetriever.ts (Line 39) & TaxPortfolioChunker.ts (Line 22)                       │
│    • RAG knowledge retrieval defaulted to [TAX_RETURN_2024, TAX_RETURN_2025] (Taylor)    │
│    • Tax Co-Pilot & Strategist prompt injections indexed synthetic demo returns in Live │
│                                                                                          │
│ 3. ProactiveTaxChatDrawer.tsx (Line 108) & /api/v1/tax/chat/route.ts                     │
│    • Called getSharedRagContext(promptText) without passing live user tax returns       │
│    • AI Copilot generated recommendations grounded in synthetic demo figures            │
│                                                                                          │
│ 4. TaxContext.tsx (Lines 98–101 & 109–111)                                               │
```

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|    • activeReturn2024 & activeReturn2025 spread DEMO_TAX_RETURN_2024/2025 in L
ive Mode |
|    • Missing years in user uploads silently inherited synthetic demo numbers
         |
|
         |
| 5. PortfolioTaxStrategyCard.tsx (Lines 163 & 170)
         |
|    • Hardcoded '742 Evergreen Terrace' directly into passive loss strategy car
ds       |
|
         |
| 6. TaxMemoryInspectorModal.tsx (Line 17)
         |
|    • Defaulted facts to INITIAL_USER_TAX_FACTS from seedData.ts
         |
|    • Displayed 'Jordan Taylor @ Pacific Cloud Systems' & 'Alex Taylor age 65'
in Live  |
|
         |
| 7. TaxPreImportReviewModal.tsx (Line 19)
         |
|    • Defaulted taxReturns prop to [DEMO_TAX_RETURN_2024, DEMO_TAX_RETURN_2025]
           |
|
         |
| 8. LifecyclePlanningEngine.ts (Line 14 & 31)
         |
|    • Default parameter defaulted to DEMO_TAX_RETURN_2025
         |
|    • earnerEmployer fell back to earnerSpouse.employerName ('Pacific Cloud Sys
tems')   |
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The Root-Cause Solution: Airtight Zero-Bleed Tenant Partitioning
In v7.1, we institute a strict architectural firewall between Demo Mode and Live
 User Mode:
1. **Isolated Profile Construction:** `resolveLiveTaxpayerProfile()` will **neve
r** spread `DEMO_TAX_FILING_PROFILE_2024_2025`. In Live Mode, it starts from a c
lean `createEmptyTaxpayerProfile()`, extracting 100% of its data from authentica
ted user metadata and parsed tax documents.
2. **Accurate Employer Resolution:** For Ivan Wong, `employerName` strictly reso
lves to **`ViaPath Technologies inc.`** (corroborated by live W-2 files and Fide
lity 401(k) accounts).
3. **Dynamic Schedule E Rental Resolution:** `PortfolioTaxStrategyCard` dynamica
lly retrieves the real property address from `scheduleE.rentalProperties[0].addr
ess` or outputs `Schedule E rental real estate property`, completely eliminating
 `742 Evergreen Terrace`.
4. **Live RAG Context Injection:** `getSharedRagContext()` and `/api/v1/tax/chat
/route.ts` will accept `liveTaxReturns`. When the user is in Live Mode, RAG stri
ctly chunks and retrieves the authenticated user's actual returns, ensuring zero
 demo data leaks into AI prompt context.
5. **No Demo Fallback in Live Mode:** In `TaxContext.tsx`, missing return years
in Live Mode will return `null` or empty models, never spreading synthetic demo
data.

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2. Universal BunBau Canonical Container UX Standard

Every page, container, card, and modal in the application will strictly implement the canonical 4-pillar BunBau UX standard:

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|                                BUNBAU CONTAINER UX STANDARD                              |
├──────────────────────────────────────────────────────────────────────────────────────────┤
|  [Icon] Container Title                                  [((i)) Info] [Chart|Table]      |
|  Subtitle / Contextual Taxpayer Labels                   [A- 100% A+] [Copy] [Expand]    |
├──────────────────────────────────────────────────────────────────────────────────────────┤
|                                                                                          |
|  MODE 1: Visual Interactive Graphics (Chart Mode)                                        |
|  • Side-by-side comparative bar charts, timeline milestone scrubbers, or gauges          |
|                                                                                          |
|  MODE 2: Structured Tabular Data (Table Mode)                                            |
|  • Searchable, sortable SmartTable with line-item breakdowns and variance columns        |
|                                                                                          |
├──────────────────────────────────────────────────────────────────────────────────────────┤
|  ACTIONABLE GUIDANCE & STRATEGY CARDS:                                                   |
|  • Statutory IRC citations, execution deadlines, and fiduciary action plans              |
|  • Dedicated [Copy Action Plan] button with instant 2-second "Copied!" toast feedback   |
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```

The 4 Canonical Container Controls:
1. **Information Explainer (`((i))`):**
   - Clickable icon opening a high-clarity popover/modal detailing:
     - The financial/tax metric definition.
     - Statutory IRC / Treasury Regulation citations (IRC §62, §469, §1212, §6654, §2210, SECURE 2.0 §107, Medicare rules).
     - CFA/CPA fiduciary methodology and planning rules.
2. **Visual Chart vs. Tabular Numbers Toggle:**
   - Seamless toggling between intuitive visual graphics (bar charts, milestone cards, threshold gauges) and structured numerical tables (SmartTable, line-item matrices).
3. **One-Click Action Item Copy (`Copy to Clipboard`):**
   - Integrated with `navigator.clipboard.writeText` and a 2-second visual confirmation (`Check` icon + "Copied!").
   - Copies clean, formatted Markdown text of fiduciary action plans, milestone dates, and strategy execution steps.
4. **Canonical Accessibility Zoom Bar:**
   - Powered by `AccessibilityZoomBar` (`[-] [100%] [+]` and reset) and responsive scale styling.

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3. Container-by-Container Standardization Specification

Container / Module	Info Icon Explainer `((i))`	Chart vs. Tabular Toggle	One-Click Copy Option	Data Isolation Fix
Overview KPI Cards (`TaxExecutiveSummaryCards.tsx`)	Popover on AGI (IRC §62), Total Tax, Effective Rate, and Loss Shields	Toggle between numeric cards and comparative Bar Chart (2025 vs 2024)	Button to copy executive KPI summary	Zero demo fallback; strictly latest vs prior live returns
Income & Tax Trends (`MultiYearTrendCard.tsx`)	Popover on income progression & bracket thresholds	Progression Bar Chart vs. SmartTable	Copy table data / CSV export	Powered by `AppletContainer` with live returns
Itemized Deductions (`MultiYearDeductionsCard.tsx`)	Popover on Schedule A deductions & standard deduction	Deduction breakdown bars vs. SmartTable	Copy table data / CSV export	Powered by `AppletContainer` with live returns
Strategic Lifecycle Roadmap (`TaxLifecycleTimeline.tsx`)	Popover on CFA/CPA milestone sequencing (Ages 62–75)	Visual timeline scrubber vs. Chronological Roadmap Table	Container "Copy All Actions" + Per-card "Copy Action Item"	**ViaPath Technologies inc.** resolved for Ivan Wong; zero Pacific Cloud residue
Social Security Planner (`SocialSecurityPlannerCard.tsx`)	Popover on SSA earnings test ($23,400) & delayed credits	Payout Bar Graph (62 vs 67 vs 70) vs. Payout Table	"Copy Claiming Strategy" button	Earner W-2 wages from live return
Medicare Health Plan (`MedicareHealthPlanCard.tsx`)	Popover on IEP window, Part B delay, Part D creditable coverage	3-Column scenario cards vs. Coordination Matrix Table	"Copy Coordination Plan" button	**ViaPath Technologies inc.** as group employer; zero Pacific Cloud
IRMAA Cliff Monitor (`IrmaaCliffMonitorCard.tsx`)	Popover on IRMAA 2-year lookback & MAGI tiers	Visual threshold gauge vs. Surcharge Tier Table	"Copy IRMAA Guidance" button	Actual MAGI from live return
RMD & Roth Conversion (`RmdRothConversionCard.tsx`)	Popover on SECURE 2.0 Age 75 RMD & 10-Yr Tax Valley	3-Pillar visual cards vs. Depletion Schedule Table	"Copy Conversion Plan" button	Uses live taxpayer ages and pre-tax balances
9-Year Longitudinal Audit (`LongitudinalComparisonCard.tsx`)	Popover on 9-year audit methodology and CAGR math	Matrix Table view vs. Macro Trends Visual Charts	"Copy Audit Summary" button	Pure live returns; zero synthetic mock rows
Line-Item Inspector Modal (in `LongitudinalComparisonCard.tsx`)	Popover on line item & IRS Form reference	9-Year numeric grid vs. 9-Year Historical Trend Chart	"Copy Line Item Analysis" button	Dynamic PDF deep links to exact page & line
Portfolio Tax Strategy (`PortfolioTaxStrategyCard.tsx`)	Popover on Schedule B/D/E cross-asset loss shielding	4-Card strategy layout vs. Loss Shield Waterfall Chart	**Dedicated "Copy Action Plan" on all 4 cards**	Purged `742 Evergreen Terrace`; dynamically loads Schedule E property
Persistent User Tax Memory (`TaxMemoryInspectorModal.tsx`)	Popover on RAG persistent factual grounding	Card list vs. Category table	"Copy Taxpayer Facts" button	Purged `INITIAL_USER_TAX_FACTS`; displays live user facts only
PDF Document Viewer (`BunBauDocumentViewer.tsx`)	Document Info modal `((i))` with file metadata	React Portal Detached Window (`z-[100]`) vs. Embedded	Copy citation / page reference	Clean display of user's uploaded tax PDFs

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4. Multi-Phase Implementation Plan

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BUNBAU v7.1 EXECUTION PHASES

Phase 1: Airtight Zero-Bleed Data Partitioning & W-2 Employer Resolution

- Refactor resolveLiveTaxpayerProfile(): create clean profile without demo
- Set Ivan Wong employerName = 'ViaPath Technologies inc.'
- Purge hardcoded 'Pacific Cloud Systems Inc.' & '742 Evergreen Terrace'
- Update TaxContext.tsx: eliminate demo spreads in Live Mode
- Update HybridRetriever & chunkTaxReturnsAndPortfolio for live returns only

Phase 2: Overview KPI Cards Standard UX (Chart/Table Toggle, Info, Copy)

- Add Info popover ((i)) on AGI, Total Tax, Effective Rate, Loss Shields
- Add Visual Bar Chart toggle (2025 vs 2024 comparison)
- Add Executive KPI Summary Copy button with 'Copied!' toast feedback

Phase 3: Strategic Lifecycle Roadmap & Forward-Looking Cards UX Standardization

- TaxLifecycleTimeline: Info icon, Chart/Table toggle, Copy All Actions
- Dedicated 'Copy Fiduciary Action Item' button on each milestone card
- Standardize Social Security, Medicare, IRMAA, and RMD/Roth cards

Phase 4: Multi-Year Longitudinal Audit & Line-Item Inspector Modal

- LongitudinalComparisonCard: Info icon, Macro Chart toggle, Copy Summary
- Line-Item Inspector Modal: 9-year trend chart toggle & Copy Analysis button

Phase 5: Active Portfolio Tax Drag & Loss Shield Strategy

- PortfolioTaxStrategyCard: Info popover & Loss Shield Waterfall chart toggle
- Dedicated 'Copy Action Plan' button on all 4 strategy cards
- Dynamic Schedule E rental address (never hardcoded 742 Evergreen)

Phase 6: Persistent Tax Memory & AI Copilot Grounding Fix

- TaxMemoryInspectorModal: eliminate INITIAL_USER_TAX_FACTS in Live Mode
- Wire ProactiveTaxChatDrawer to pass live tax returns to RAG retrie

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         |
|  Phase 7: Version Synchronization, Regression Testing & Production Build
         |
|           • package.json: 2.7.13, AboutModal: 2.7.13 (Build 128), Arch: v2.9.6
         |
|           • Authored comprehensive unit tests for data isolation & standard UX
         |
|           • 100% green test execution (625+ tests) & verified Next.js 15 build
         |
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5. Verification & Acceptance Criteria

1. **Zero Demo Data Leakage:**
   - In Live Mode, search across DOM, RAG prompt contexts, and state confirms **zero** occurrences of `Alex Taylor`, `Jordan Taylor`, `Pacific Cloud Systems Inc.`, `742 Evergreen Terrace`, or `San Francisco Wealth & Tax Advisory`.
2. **ViaPath Technologies inc. Verification:**
   - Milestone Year 2027 reads: *"Coordinate with Ivan Wong’s ViaPath Technologies inc. employer health insurance."*
   - Fiduciary Action Item reads: *"Delay Part B & Part D without penalty if ViaPath Technologies inc. group coverage remains active."*
   - Medicare coordination card lists `ViaPath Technologies inc.` as Ivan Wong's employer.
3. **Universal Container UX Consistency:**
   - Every single container features an Info explainer `((i))`.
   - All analytical containers feature a seamless toggle between Visual Chart and Tabular Numbers.
   - Every action recommendation card features a dedicated Copy button with immediate visual feedback.
   - All containers comply with accessibility scaling via `AccessibilityZoomBar`.
4. **Complete Test Suite & Production Build:**
   - 106+ test suites passing with 100% green status.
   - Next.js 15 production build compiles with zero errors across all 16 static and dynamic routes.

Please review this comprehensive plan. Once you click **Proceed** or reply with approval, I will execute all phases immediately.